

# CENG 589 Term Project

## Parameter Study for Mesh Approximation and Deformation using Least-Squares

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### Abstract

This work presents a unified and flexible framework for mesh approximation and deformation using a least-squares formulation based on the Laplacian matrix and differential coordinates of a 3D mesh. By introducing a smoothing parameter  $\lambda$ , the proposed approach enables continuous control over the trade-off between geometric fidelity and smoothness, facilitating applications such as mesh simplification, denoising, and shape-preserving deformation. Through parameter studies, the influence of control point density and the smoothing factor on mesh quality is analyzed using a cow model. The method is extended to support localized deformation by modifying the influence of differential coordinates in selected regions via a scale matrix. Additionally, alternative deformation functions—linear and hyperbolic tangent—are explored to reduce artifacts and ensure more natural deformations. Results demonstrate that the framework offers robust, interpretable, and tunable behavior suitable for practical mesh editing tasks, implemented efficiently using the Open3D library.

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## 1. Introduction

In digital geometry processing, mesh approximation and deformation are fundamental tasks with broad applications ranging from geometry editing to simulation and shape analysis. Effective methods for mesh manipulation must balance competing goals such as preserving geometric fidelity, maintaining smoothness, and adhering to user-defined constraints. This work explores a unified framework for mesh approximation and deformation that provides fine control over these trade-offs through tunable parameters.

The methodology is grounded in solving linear least-squares systems involving the Laplacian matrix of the mesh. Two prominent formulations from the literature are first reviewed: one prioritizing smoothness by minimizing the Laplacian directly, and the other emphasizing shape fidelity by aligning the mesh with its differential coordinates. A hybrid formulation is then proposed, introducing a smoothing parameter to interpolate between these two objectives. This formulation not only allows approximating the original mesh from a sparse set of control points, but also enables denoising and controlled simplification.

To analyze the behavior of the proposed approach, I conduct several parameter studies involving control point density and the smoothing factor. These experiments reveal how geometric characteristics such as volume and local detail preservation vary with parameter changes. Furthermore, the same formulation is extended to support localized mesh deformation. A scale-aware adjustment is introduced in the Laplacian term to produce smooth deformations within a region of interest, while preserving the global structure of the shape. Alternative deformation functions, including a hyperbolic tangent weighting scheme, are also explored to reduce artifacts and improve user control.

The remainder of this article is structured as follows: Section 2 details the mesh approximation methodology and its effects under various settings. Section 3 builds upon this foundation to propose a novel approach to mesh deformation, analyzing how scale and weighting affect the results. Experimental evaluations and visualizations are provided throughout to illustrate the practical impact of each parameter and formulation.

In this project, I used an SDK called Open3D which is an open-source Python library for 3D data processing and visualization.

## 2. Mesh Approximation

In this section, the methodology for the mesh approximation is described, and the application of it to several examples for the parameter studies is presented.

### 2.1. Methodology

To express the mesh in the least-squares sense, [1] uses the following linear system

$$\bar{x} = \arg \min_x \left( \|Lx\|^2 + \sum_{j \in C} w_j^2 |x_j - c_j|^2 \right) \quad (1)$$

while [2] uses the following linear system

$$\bar{x} = \arg \min_x \left( \|Lx - \delta^{(x)}\|^2 + \sum_{j \in C} w_j^2 |x_j - c_j|^2 \right) \quad (2)$$

where  $x$  is a vector containing vertex coordinates of the mesh,  $L$  is the Laplacian matrix of the mesh,  $C$  is the set of indices of control points,  $c$  is a vector of coordinates of the control points, and  $w$  is the weight parameter used to tweak the importance of the control points.  $\delta^{(x)}$  is the vector of differential coordinates of the vertices which is only present in the second equation. It is defined as

$$\delta_i = (\delta_i^x, \delta_i^y, \delta_i^z) = v_i - \frac{1}{d_i} \sum_{j \in N(i)} v_j \quad (3)$$

Although the equations (1) and (2) look quite similar, they represent different formulations of optimization objectives. The difference comes from the Laplacian terms of the equations. In the first equation, it is attempted to flatten or smooth the mesh by minimizing the  $Lx$  term, while in the second equation, it is attempted to match the original differential geometry by minimizing the  $Lx - \delta^x$  term. Therefore, the output mesh of the second equation resembles the original local structure, while the output mesh of the first equation is smoother, more regular, and less detailed.

I combined these two equations to obtain

$$\bar{x} = \arg \min_x \left( \lambda \|Lx - \delta^{(x)}\|^2 + (1 - \lambda) \|Lx\|^2 + \sum_{j \in C} w_j^2 |x_j - c_j|^2 \right) \quad (4)$$

where  $\lambda \in [0, 1]$  is the smoothing parameter.  $\lambda$  controls the trade-off between smoothness and original shape matching. While  $\lambda$  is decreasing, the resulting mesh becomes smoother at the cost of losing fidelity. By this equation, it is aimed to gain insight about shape matching and smoothing. The equation is solved for  $x$  by taking the derivative of the least-squares function and equating it to 0. Then, we get

$$2\lambda(L^2x - L\delta^{(x)}) + 2(1 - \lambda)(L^2x) + 2 \sum_{j \in C} w_j^2 (x_j - c_j) = 0 \quad (5)$$

Arranging this equation to get

$$L^2x - \lambda L\delta^{(x)} + \sum_{j \in C} w_j^2 (x_j - c_j) = 0 \quad (6)$$

I use a direct method for solving this sparse linear least-squares system.

As seen in these equations, we need some control points to approximate the mesh. For this reason, I use farthest point sampling (FPS) to find shape extremities. These points become the control points for the creation of the new mesh. The FPS algorithm needs shortest geodesic distances between vertices, so to find the shortest distance between two vertices on the surface, Dijkstra's shortest path algorithm is implemented.

## 2.2. Results and Findings

Here, the mesh approximation methodology will be applied to several examples to investigate the effect of the parameters previously defined.

In the first example, the effect of the number of control points on the resultant mesh will be shown. A cow model composed of 2903 vertices and 5804 faces is approximated by equation (6) for different numbers of control points as shown in Figure 1.  $\lambda$  is taken as for all the results. As the number of control points increases, the similarity between the resultant and the original mesh increases. For only 64 or 128 control points compared to 2903 vertices, a mesh quite similar to the original mesh can be generated.

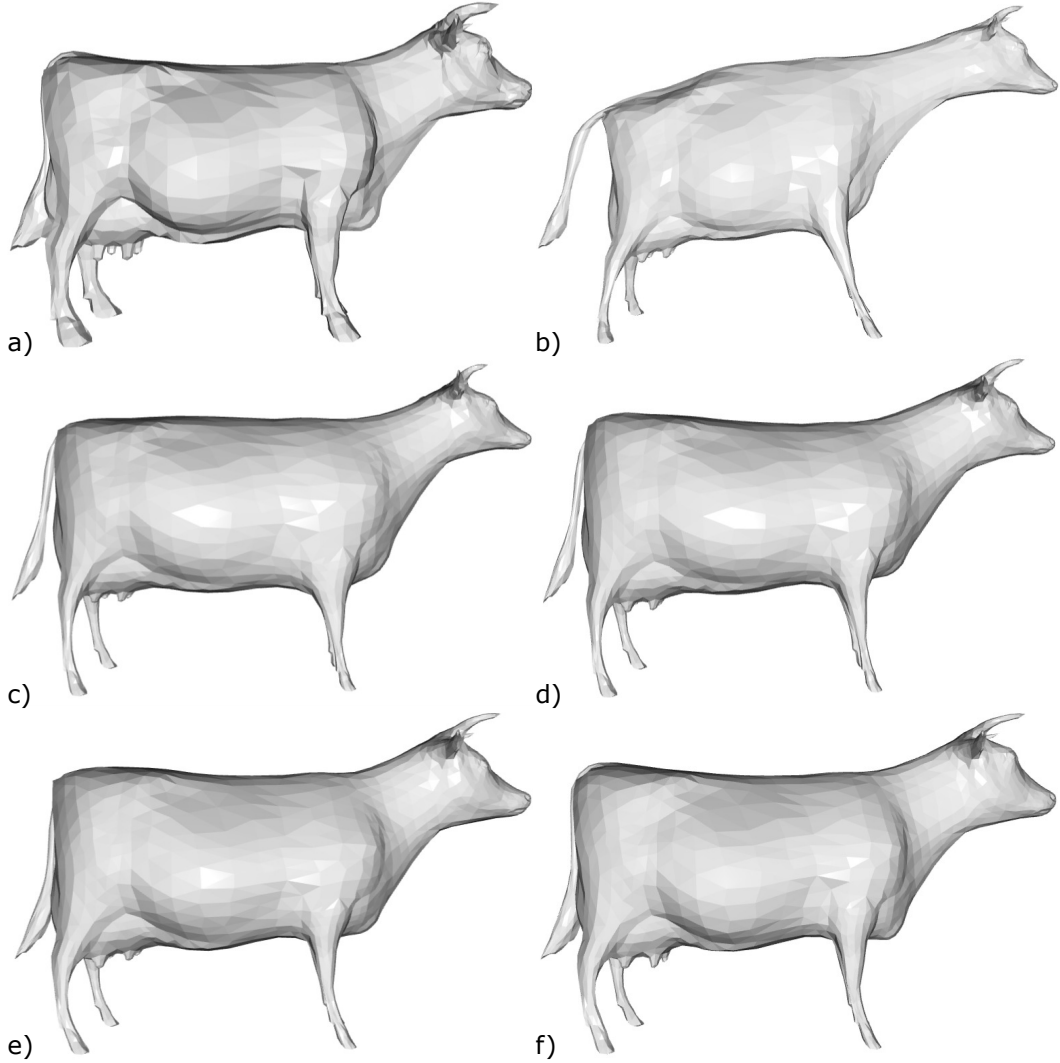


Figure 1: Parameter study for the number of control points. a) Original shape, b) 8, c) 16, d) 32, e) 64, f) 128 control points

To investigate the effect of the number of control points quantitatively, the volumes of the resultant meshes are compared. [3] argues that the volume of the mesh is useful in this context. The volumes of the resultant meshes with respect to the number of control points are plotted, and as shown in Figure 2, as the number of control points increases, the volume of the resultant mesh converges to the volume of the original mesh.

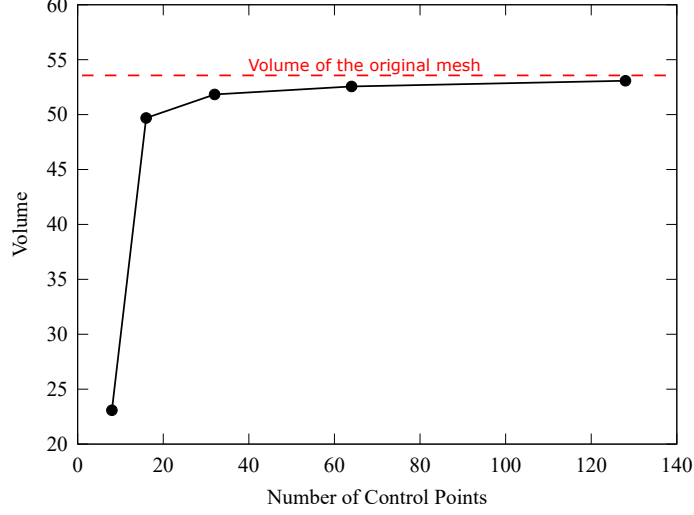


Figure 2: Volume of the resultant mesh for different number of control points.

As mentioned previously,  $\lambda$  controls the trade-off between smoothness and original shape matching. While  $\lambda$  is decreasing, the resulting mesh becomes smoother at the cost of losing fidelity. Therefore, to investigate the effect of  $\lambda$  on the resultant mesh, the same cow model from the previous example is approximated by equation (6) for different  $\lambda$  values as shown in Figure 3. The number of control points is 32 for all the results. As the value of  $\lambda$  increases, the similarity between the resultant and the original mesh increases while the resulting meshes with lower  $\lambda$  values are smoother.

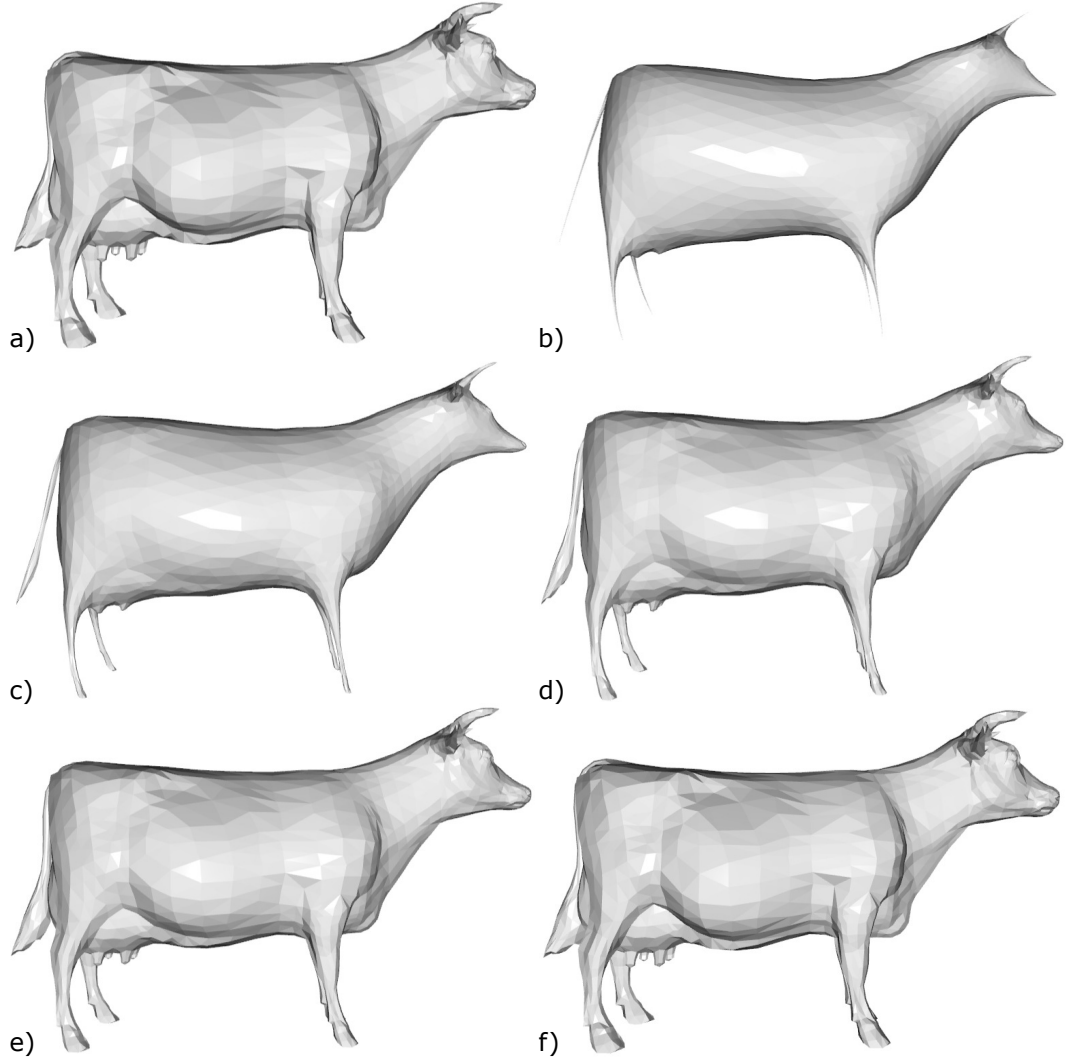


Figure 3: Parameter study for  $\lambda$ . a) Original shape, b)  $\lambda = 0$ , c)  $\lambda = 0.25$ , d)  $\lambda = 0.5$ , e)  $\lambda = 0.75$ , f)  $\lambda = 1$

To investigate the effect of  $\lambda$  quantitatively, the volumes of the resultant meshes are compared. The volumes of the resultant meshes with respect to  $\lambda$  values are plotted, and as shown in Figure 4, as  $\lambda$  increases, the volume of the resultant mesh rapidly increases. The volume of the resultant mesh with  $\lambda = 1$  is almost the same as the volume of the original mesh.

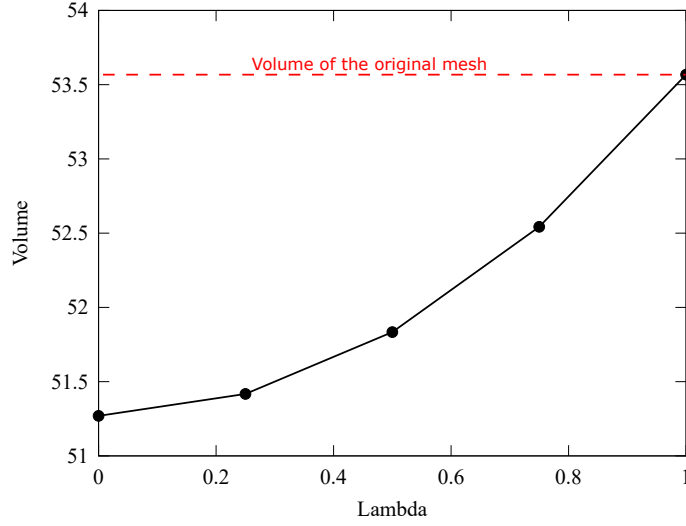


Figure 4: Volume of the resultant mesh for different  $\lambda$  values.

As shown previously, while the  $\lambda$  value is decreasing, the resultant mesh becomes smoother. Therefore, we can use this effect to denoise noisy meshes. A duck mesh is corrupted by Gaussian noise of 0.2. Then, this noisy mesh is approximated by equation (6) for different  $\lambda$  values as shown in Figure 5. The number of control points is 128 for all the results. While the  $\lambda$  value is decreasing, the resulting mesh becomes smoother; for  $\lambda$  values 0.25 and 0, the mesh is almost denoised completely.

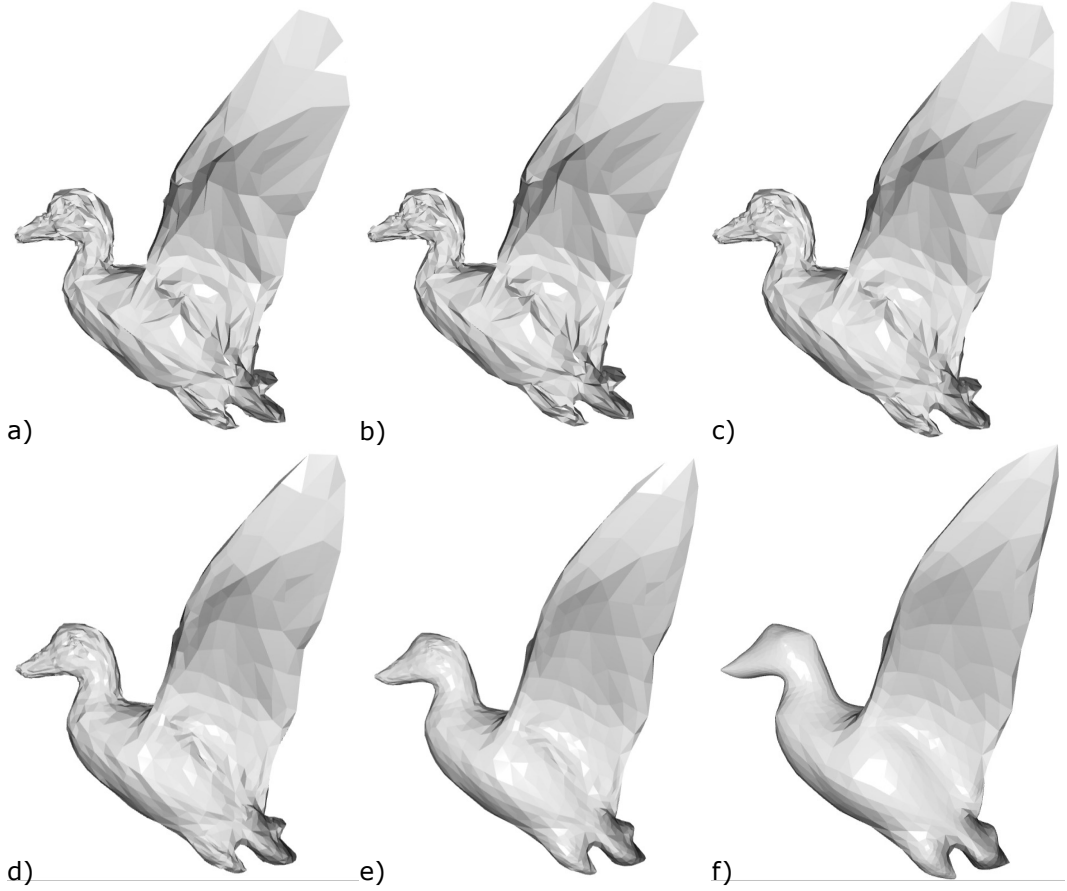


Figure 5: Smoothing of the noisy model. a) Original shape, b)  $\lambda = 1$ , c)  $\lambda = 0.75$ , d)  $\lambda = 0.5$ , e)  $\lambda = 0.25$ , f)  $\lambda = 0$

### 3. Deformation

In this section, the methodology for the deformation is described, and the application of it to several examples for the parameter studies is presented.

#### 3.1. Methodology

As mentioned in the previous section, in equation (4), it is attempted to flatten or smooth the mesh by minimizing the  $Lx$  term while it is tried to match the original differential geometry by minimizing the  $Lx - \delta^x$  term. For deformation, it is desirable to make the part that is moved deform smoothly while making the remaining part match the original shape. Therefore, in the light of equation (6), I propose a minimization equation in the following form

$$\bar{x} = \arg \min_x \left( \|Lx - K^{(x)}\delta^{(x)}\|^2 + \sum_{j \in C} w_j^2 |x_j - c_j|^2 \right) \quad (7)$$

where I introduce a diagonal scale matrix  $K^x$

$$K_{i,j}^{(x)} = \begin{cases} 0 & \text{if } i \neq j \\ s & \text{if } i = j \text{ and } i, j \in D \\ 1 & \text{otherwise} \end{cases} \quad (8)$$

where  $s \in [0, 1]$  is a scale factor and  $D$  is a set composed of the indices of the moved vertices. By that logic, I decrease the effect of differential coordinates for the deformed part to smooth the deformation. Since the deformed part does not stay the same anyway, we can compromise shape matching for smooth deformation. However, smoothing is not applied to other parts to keep the shape as matching to the original shape as possible.

For the Deformation part, user selects a handle vertex, and the deformation is applied to the points near the handle by a previously defined threshold value.

The deformation is applied in a way that the points closer to the handle deform more while the points farther from the handle deform less. The variation of the deformation of points is linearly dependent on the distance to the handle. For a deformation vector of handle vertex  $\Delta_h = (\Delta_x, \Delta_y, \Delta_z)$ , the deformation vector for other points is defined as

$$\Delta_i = \begin{cases} \Delta_h \times t & \text{if } i \in C \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where  $t = 1 - d/d_{max}$ . Here,  $d$  is the distance of the point  $i$  to the handle vertex, and  $d_{max}$  is the distance of the farthest point in the deformed region to the handle vertex.

#### 3.2. Results and Findings

Here, the deformation methodology will be applied to several examples to investigate the effect of the parameters previously defined.

As mentioned previously, while the scale factor  $s$  defined in equation (8) is decreasing, the effect of differential coordinates for the deformed part is decreasing, resulting in smoother deformation. Therefore, to investigate the effect of the scale factor  $s$  on



the deformed mesh, the same cow model from the previous examples undergoes deformation by equation (7) for different  $s$  values as shown in Figure 6. The tail of the cow is the region of interest (ROI) and the tip of the tail is the handle. The handle is moved, and the deformation vector is applied on the ROI according to the equation (9), and the body deforms accordingly. As  $s$  value is decreasing, the tail of the cow, which is the ROI, becomes smoother. However, the other parts remain almost the same since smoothing is applied only to the ROI.

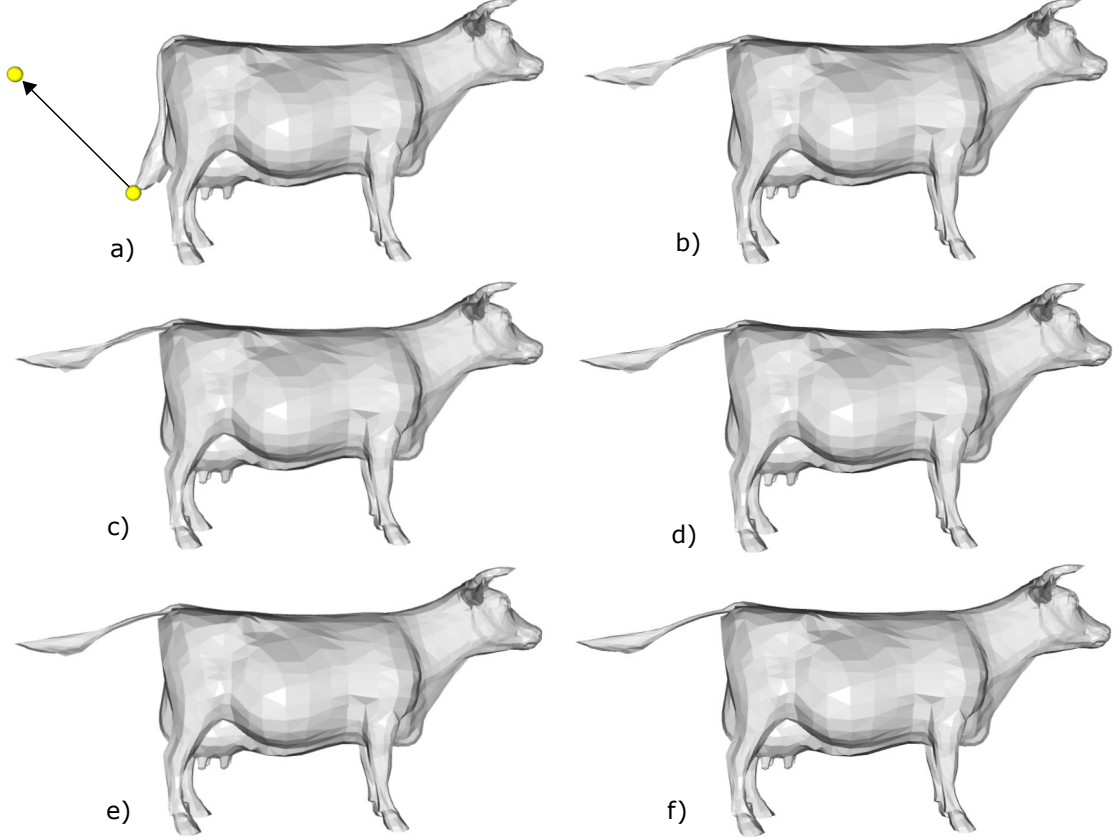


Figure 6: Parameter study for  $s$ . a) Handle and the direction of the deformation, b)  $s = 1$ , c)  $s = 0.75$ , d)  $s = 0.5$ , e)  $s = 0.25$ , f)  $s = 0$

As described previously, the weight parameter  $w$  shown in equation (7) is used to tweak the importance of the control points. Therefore, if we want the points outside the region of interest also to deform according to the applied deformation, we can decrease the weight for those points. To investigate the effect of the weight parameter  $w$  outside the ROI on the deformed mesh, the same cow model from the previous examples undergoes deformation for different  $w$  values as shown in Figure 7.  $s$  is taken as 0.5 for all the results. The deformation is applied in the same way described previously. As  $w$  value for the outside of ROI is decreasing, the body becomes less rigid, resulting in deformation outside the ROI. However, the ROI, which is the tail of the cow, is not affected, deforming in a similar way for all the cases since the  $w$  value for the points in ROI is 1 for all the cases.

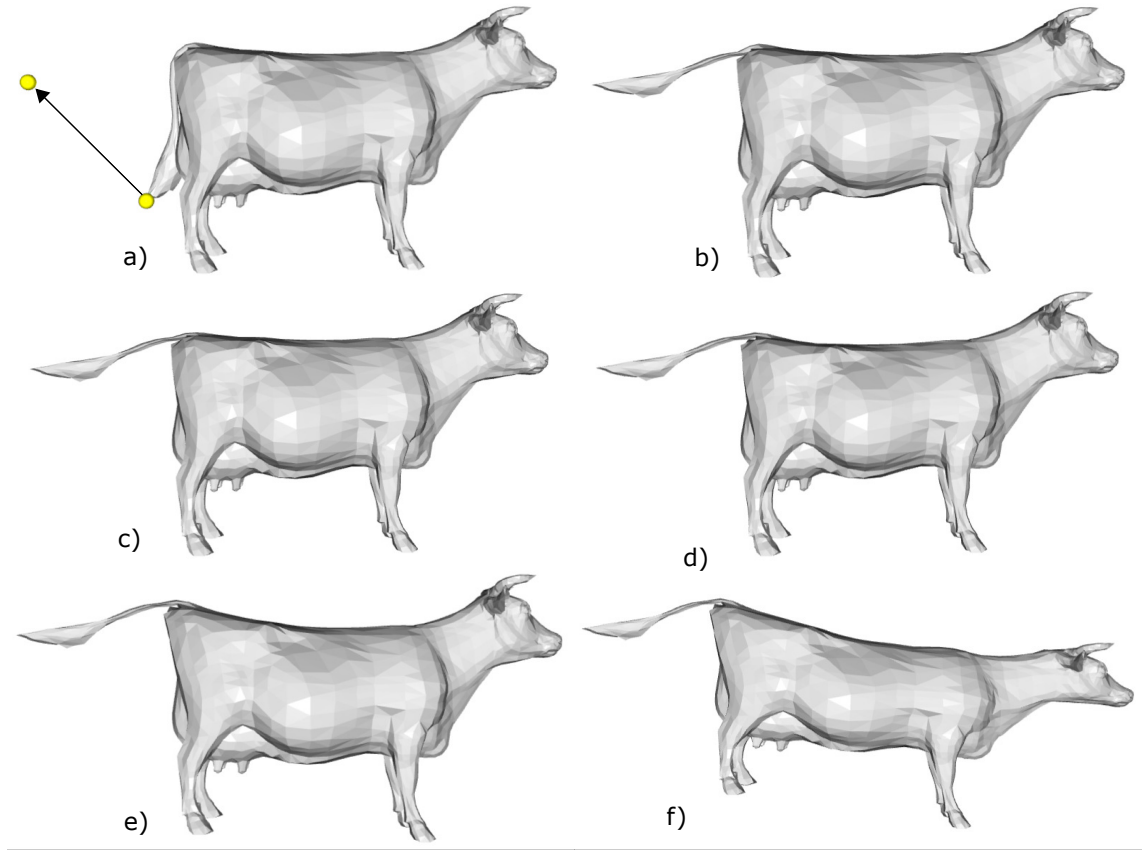


Figure 7: Parameter study for  $w$  a) Handle and the direction of the deformation, b)  $w = 1$ , c)  $w = 0.1$ , d)  $w = 0.01$ , e)  $w = 0.001$ , f)  $w = 0$

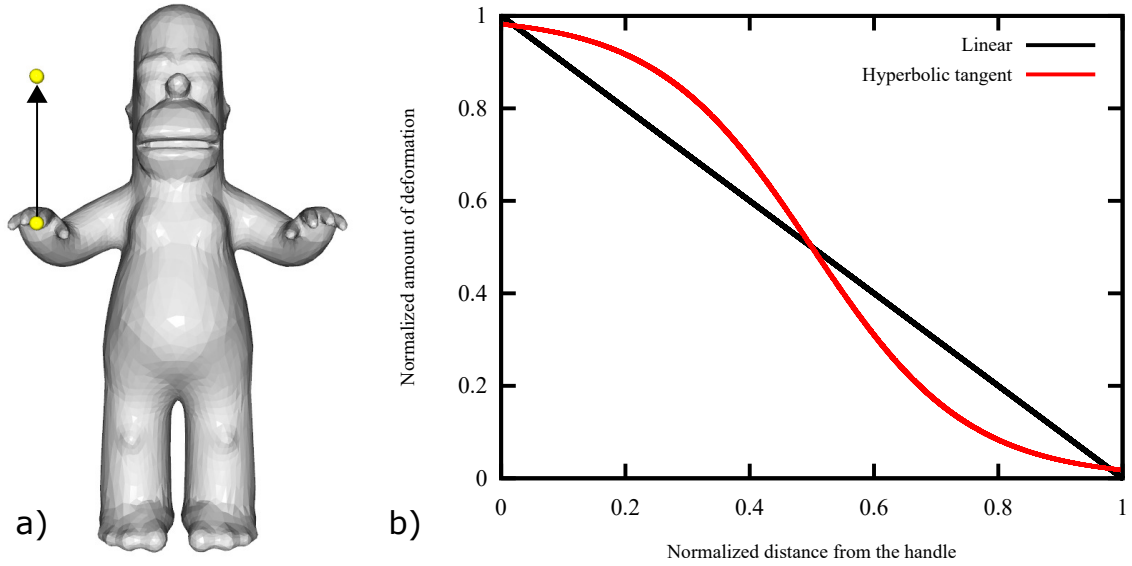


Figure 8: a) Handle and the direction of the deformation. b) Deformation functions.

Until now, we have used the linear variation of the deformation of points in ROI dependent on the distance to the handle defined in equation (9). However, this approach may lead to undesired results for certain shapes at locations near and away from the handle because of the sharp variation of deformation. Therefore, I used a hyperbolic

tangent function instead of the linear one to investigate the effect of the variation of the deformation. For a deformation vector of handle vertex  $\Delta_h = (\Delta_x, \Delta_y, \Delta_z)$ , the deformation vector for other points is defined by equation (9) where

$$t = \left[ 1 - \tanh \left( 4 \left( \frac{d}{d_{max}} - \frac{1}{2} \right) \right) \right] / 2 \quad (10)$$

This function will be used for the deformation of the mesh shown in Figure 8. The right arm is the region of interest (ROI) and the vertex shown with a yellow ball is the handle. Also, in the same figure, the comparison of the linear and hyperbolic tangent deformation functions is demonstrated. The variation of deformation at locations near and away from the handle is lower for the hyperbolic tangent function than for the linear function.

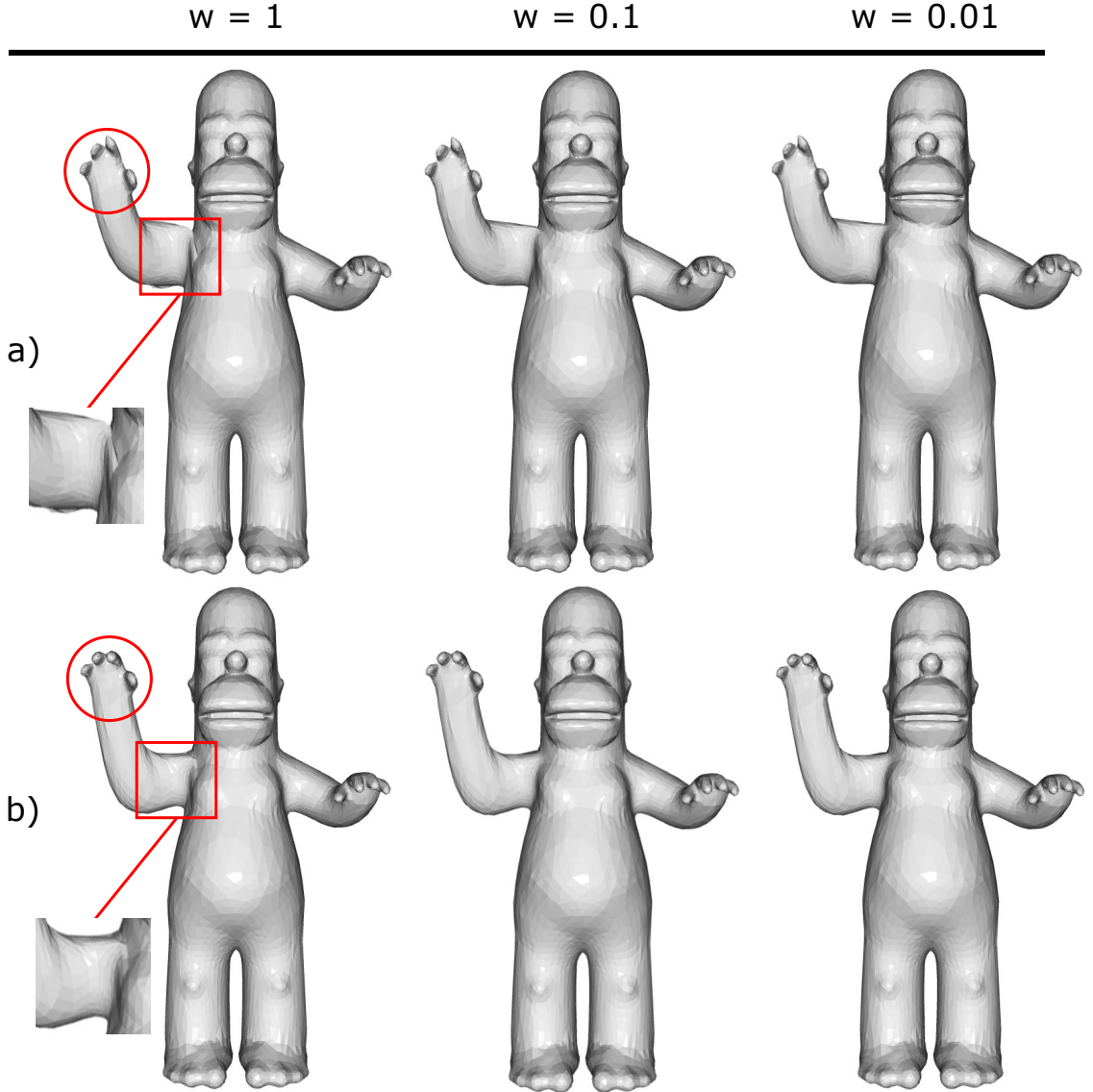


Figure 9: Parameter study for  $w$  and the deformation function a) linear, b) hyperbolic tangent

The results for different values of the weight parameter  $w$  outside ROI and the deformation function are shown in Figure 9.  $s$  is taken as 0.5 for all the results. For the linear deformation function, the hand where the handle is located is distorted

after deformation for all the values of  $w$ . Also for the  $w = 1$  case with the linear deformation function, there are observable artifacts at the armpit which is the farthest region from the handle. On the other hand, for the hyperbolic tangent deformation function, no distortion or artifacts are observed at the hand or armpit. However, in the between region, a stretching effect is observed because of the increase in the variation of deformation in that region by the hyperbolic tangent function.

#### 4. Conclusions

In this work, I presented a unified framework for mesh approximation and deformation based on a flexible least-squares formulation involving the Laplacian of the mesh and its differential coordinates. By introducing a smoothing parameter  $\lambda$ , it is possible to control the trade-off between preserving the original geometric features and achieving a smoother representation. This balance is crucial for applications such as mesh simplification, editing, and denoising.

Through parameter studies on a cow model, I demonstrated that increasing the number of control points enhances the fidelity of the approximated mesh, while decreasing the  $\lambda$  parameter results in smoother but less accurate approximations. Volume analysis confirmed that the approximation converges toward the original shape as these parameters are optimized appropriately. Additionally, the method proved effective in denoising noisy meshes by exploiting the smoothing effect of lower  $\lambda$  values.

In the second part of the study, I extended the formulation to support controlled mesh deformation. A scale-aware modification to the Laplacian term allowed selective smoothing in the region of interest (ROI), making it possible to deform localized areas while maintaining the integrity of the rest of the shape. Parameter studies involving the scale factor  $s$ , weight parameter  $w$ , and deformation functions revealed how to modulate deformation behavior across different regions of the mesh. Notably, the use of a hyperbolic tangent function instead of a linear one helped eliminate artifacts near the handle and in distant regions, resulting in more visually consistent deformations.

Overall, the proposed approach provides a versatile and computationally efficient way to approximate and deform triangle meshes with fine-grained control over smoothness and fidelity. It can be integrated into various geometry processing pipelines where mesh quality and user control are critical.

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